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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/319,618	05/18/2023	Samuel Benjamin Pollock	P36373-US-1	8596

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EXAMINER
MONTGOMERY, ANN Y

ART UNIT	PAPER NUMBER
1678	

NOTIFICATION DATE	DELIVERY MODE
06/03/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docketing-uspto-d@gene.com

Office Action Summary	Application No. 18/319,618	Applicant(s) Pollock, Samuel Benjamin	
	Examiner Ann Montgomery	Art Unit 1678	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 3/31/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 22,24,26 and 37-47 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 22,24,26 and 37-47 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 11/4/25, 4/29/25, 5/18/23.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claim 22, 24, 26 and 37-47 is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 22 recites in line 6 (subsection (c)) that the cartridge to be prepared for re-use comprises "at least one cryoprotectant".

However, claim 22 also recites in line 10 "washing the matrix in a buffer comprising the at least one cryoprotectant".

Thus it is not clear how the cartridge both comprises "at least one cryoprotectant" (in line 6) as well as comprises a matrix that is *washed* in a buffer that comprises *the same* "at least one cryoprotectant". For examination purposes, the claim will be interpreted to mean that the matrix is

washed in a buffer that comprises the same type of cryoprotectant as recited in line 6, or that comprises *a* cryoprotectant [rather than the *same* cryoprotectant recited in line 6.]

Moreover, the term "room temperature" in claim 24 renders the claim indefinite since the term "room temperature" is not defined by the claim, the specification does not provide a standard for ascertaining what temperature or temperature range constitutes "room temperature", and one of ordinary skill in the art would not be reasonably apprised of the scope of the invention. Examiner notes that while the term "room temperature" may have been often used in the art, there is no definite meaning to "room temperature" in the art, and thus the term is unclear. Room temperature can vary depending on the setting and/or thermostat set for the room. The limitation of "room temperature" renders the claim indefinite since the meaning of the term cannot be ascertained.

The remaining claims are rejected since they depend from claim 22, which is vague for the reason set forth above, without clarifying the vagueness.

Allowable Subject Matter

Claim 22 would be allowable if rewritten or amended to overcome the rejection(s) under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), 2nd paragraph, set forth in this Office action.

Claim 24, 26 and 37-47 would be allowable if rewritten to overcome the rejection(s) under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), 2nd paragraph, set forth in this Office action and to include all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter.

It was not found in the prior art search a teaching or suggestion for a method of preparing a previously used cartridge for re-use,

wherein the **cartridge comprises:** a) a matrix, b) **detection molecules attached** to the matrix, wherein the detection molecules specifically bind to an analyte, and c) at least one cryoprotectant,

the method comprising,

after analyte has been eluted from the cartridge, **washing the matrix in 1-10% acetic acid, 1% formic acid, or 1% trifluoroacetic acid (TFA);**

washing the matrix in a buffer comprising [a] cryoprotectant; and

allowing the matrix to dry.

Examiner notes that the present application claims priority to Provisional application no. 63116575 (filed 11/20/20) which discloses the presently claimed invention in paragraphs 0005 and 0006.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Pollock et al. "Sensitive and Quantitative Detection of MHC-I Displayed Neoepitopes Using a Semiautomated Workflow and TOMAHAQ Mass Spectrometry" 2021, Mol Cell Proteomics 20, 100108. This reference discloses a cartridge washed with 1% acedic acid. To make dried cartridges, an additional wash with trehalose was made, and the cartridge dried. Reused wet cartridges were primed with 1% acetic acid, and washed with water before reuse. Page 4, first full para.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Ann Montgomery whose telephone number is (571)272-0894. The examiner can normally be reached Mon-Fri, 9-5:30 PM PST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Greg Emch can be reached at 571-272-8149. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Ann Montgomery/
Primary Examiner, Art Unit 1678